Q1)What is SQL and its Use?

Ans: SQL is a standard language for accessing and manipulating databases.SQL stands for Structured Query Language. SQL lets you access and manipulate databases.
SQL became a standard of the American National Standards Institute (ANSI) in 1986, and of the International Organization for Standardization (ISO) in 1987.
Uses of SQL are as follows:

1. Querying Data: Retrieving specific data from one or more tables using commands like SELECT.
2. Inserting Data: Adding new records to a database table with commands like INSERT.
3. Updating Data: Modifying existing records using commands like UPDATE.
4. Deleting Data: Removing records from a table with commands like DELETE.
5. Creating and Altering Tables: Defining new tables or modifying existing ones using commands like CREATE TABLE and ALTER TABLE.
6. Managing Database Structures: Handling database schemas, indexes, and constraints to ensure data integrity and optimize performance.

Q2) Describe all the different commands of SQL, mention use of it and syntax.

Ans: 1) DDL (Data Definition Language) - DDL commands are used to create and modify the structure of database objects such as databases and tables.

a) CREATE DATABASE:

Use: Creates a new database.

Syntax: CREATE DATABASE database_name;

b) CREATE TABLE:

Use: Creates a new table.

Syntax: CREATE TABLE table_name (

column1 datatype,

column2 datatype,

column3 datatype);

c) ALTER TABLE

Use: Adds, deletes, or modifies columns.

Syntax: ALTER TABLE table_name

ADD column_name datatype;

d) DROP TABLE

Use: Permanently deletes a table.

Syntax: DROP TABLE table_name;

e) TRUNCATE TABLE

Use: Deletes all rows while keeping the table structure.

Syntax: TRUNCATE TABLE table_name;

f) CREATE INDEX

Use: Creates an index to improve search performance.

Syntax:CREATE INDEX index_name

ON table_name(column_name);

g) DROP INDEX

Use: Removes an index.

Syntax: DROP INDEX index_name;

2) DML (Data Manipulation Language) - DML commands manipulate the data stored inside tables.

a) INSERT INTO

Use: Inserts new records.

Syntax: INSERT INTO table_name(column1, column2)

VALUES(value1, value2);

b) UPDATE

Use: Modifies existing records.

Syntax: UPDATE table_name

SET column_name = value

WHERE condition;

c) DELETE

Use: Deletes records.

Syntax: DELETE FROM table_name

WHERE condition;

3. DQL (Data Query Language) - DQL is mainly used to retrieve data.

a) SELECT

Use: Fetches data from one or more tables.

Syntax: SELECT column_name

FROM table_name;

Retrieve all columns:

SELECT *

FROM Student;

Common Clauses Used with SELECT:

WHERE :- Use: Filters records.

Syntax: SELECT *

FROM table_name

WHERE condition;

ORDER BY:- Use: Sorts data.

Syntax: SELECT *

FROM table_name

ORDER BY column_name ASC;

GROUP BY:- Use: Groups rows with the same values.

Syntax: SELECT column_name,

COUNT(*)

FROM table_name

GROUP BY column_name;

HAVING :- Use: Filters grouped data.

Syntax: SELECT column_name

FROM table_name

GROUP BY column_name

HAVING condition;

4. DCL (Data Control Language) - DCL commands control user access and permissions.

a) GRANT

Use: Gives privileges to users.

Syntax: GRANT privilege

ON table_name

TO user_name;

b) REVOKE

Use: Removes privileges.

Syntax: REVOKE privilege

ON table_name

FROM user_name;

5. <u>TCL (Transaction Control Language) </u>- TCL commands manage transactions so that changes can be saved or undone.

a) COMMIT

Use: Saves all changes permanently.

Syntax: COMMIT;

b) ROLLBACK

Use: Undoes changes since the last commit.

Syntax: ROLLBACK;

c) SAVEPOINT

Use: Creates a checkpoint within a transaction.

Syntax: SAVEPOINT savepoint_name;

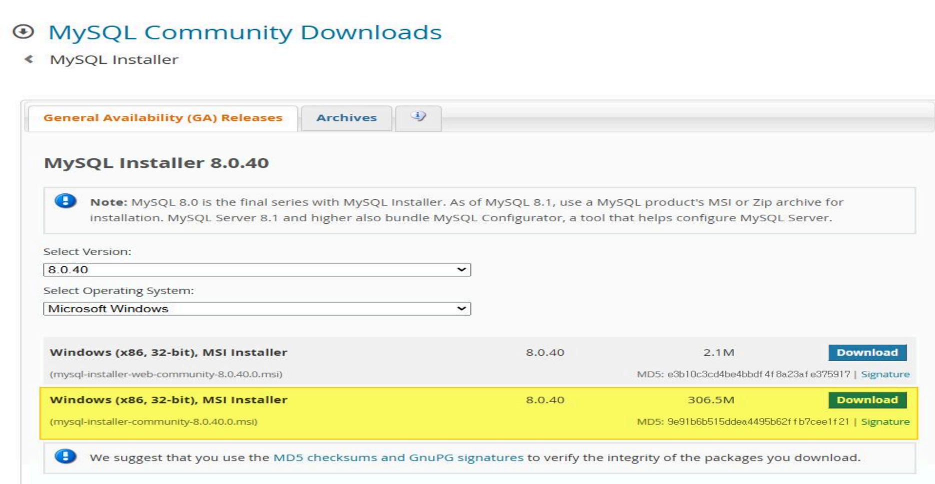
d) SET TRANSACTION

Use: Sets transaction properties.

Syntax: SET TRANSACTION;

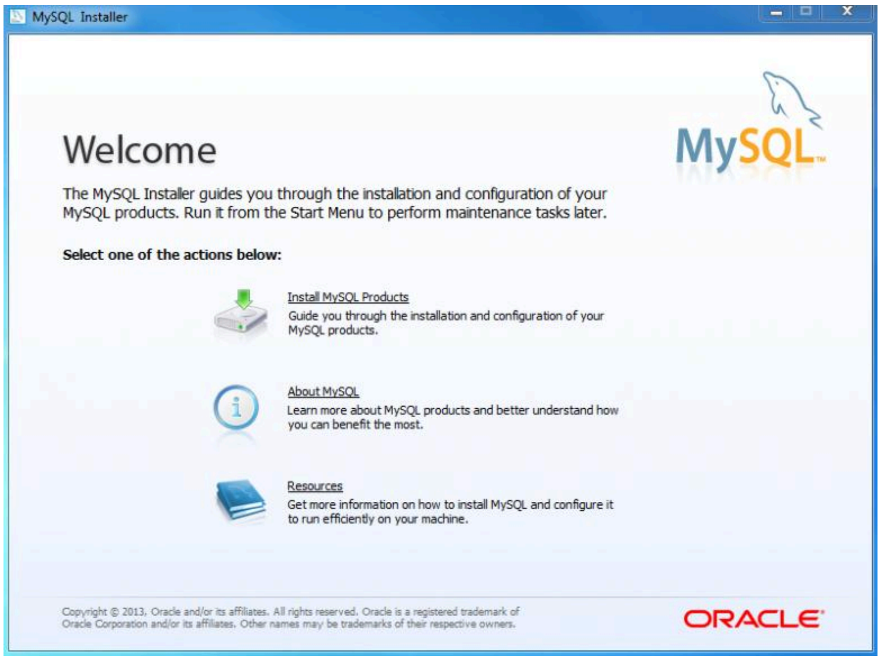
Q3) Write step by step procedure to install MYSQL.

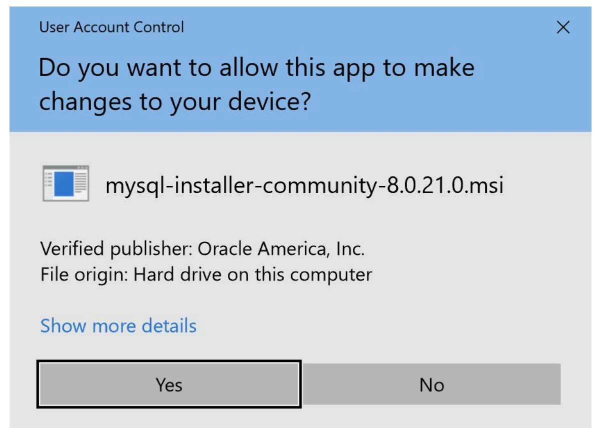
Ans: Step 1: Download MySQL Installer



1. Open your browser.
2. Go to the MySQL Community Downloads page.
3. Download MySQL Installer for Windows (.msi).
4. Choose either:
   - Web Installer (smaller download)
   - Full Installer (includes all packages)

Step 2: Run the Installer.





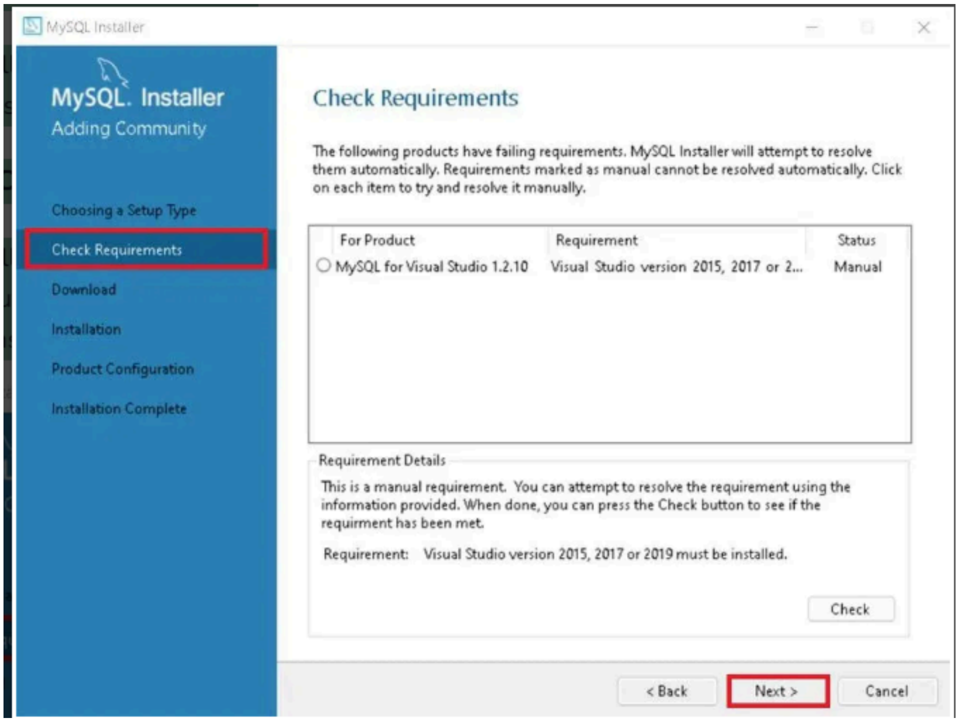
1. Double-click the downloaded .msi file.
2. If Windows asks "Do you want to allow this app to make changes?", click Yes.
3. The MySQL Installer will open.

Step 3: Choose Setup Type.

1. The installer will instruct you to choose the setup type.
2. For most users, the "Developer Default" is suitable. Click "Next" to proceed.

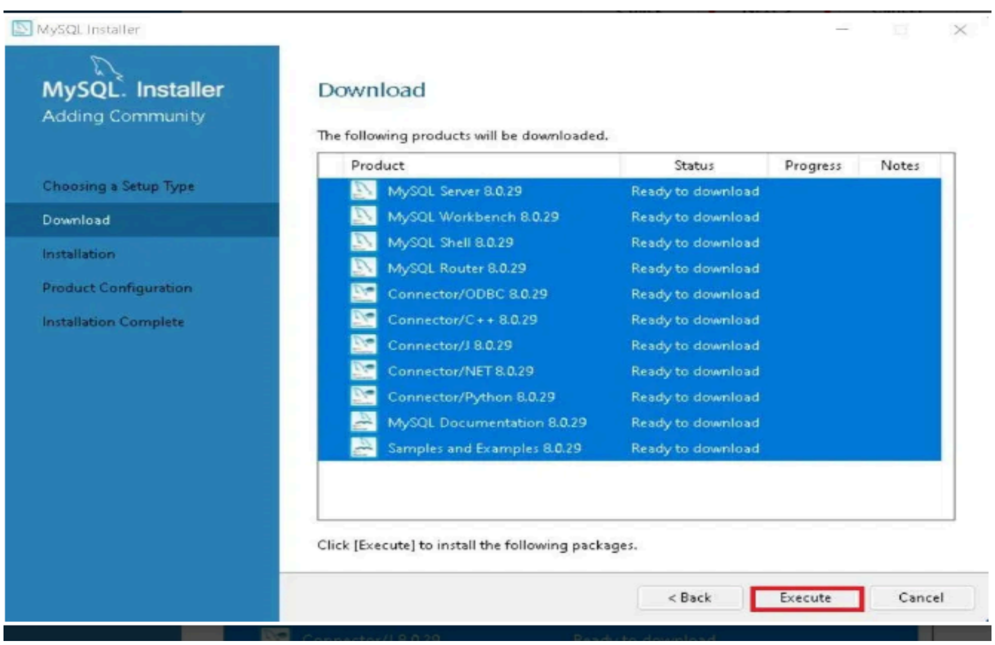
Step 4: Check Requirements

1. The installer will check for required dependencies such as the Microsoft Visual C++ Redistributable.
2. It may automatically resolve missing components, but in some cases, manual installation may be required.



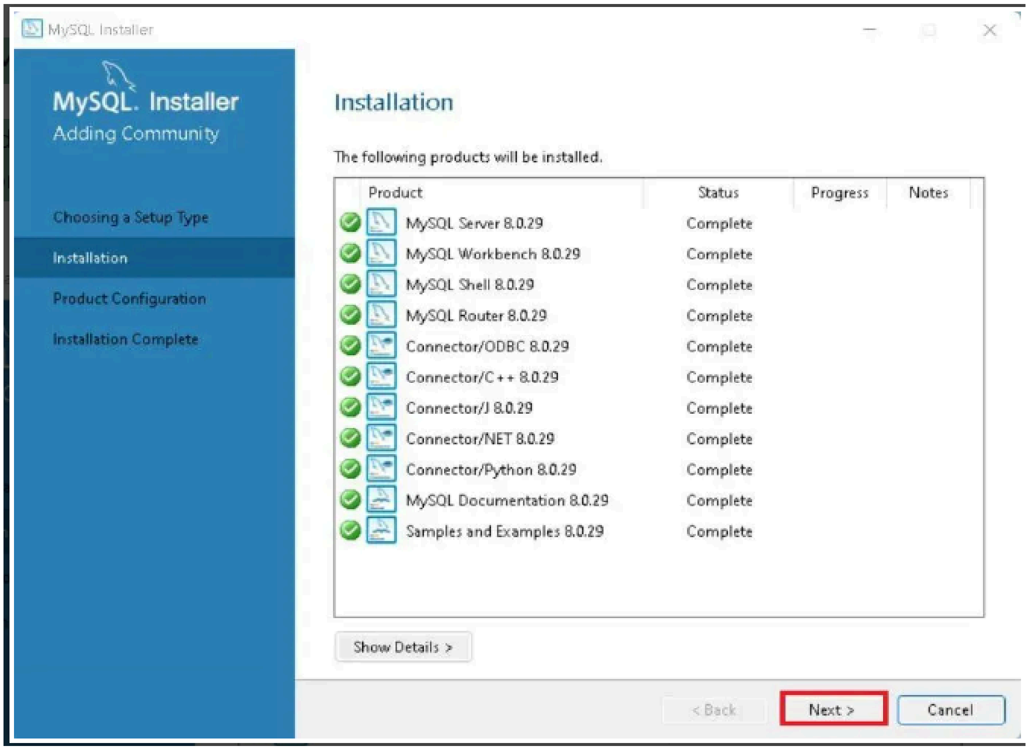
Step 5 : MySQL Downloading

1. in the download section, click "Execute" to start downloading the components you selected.
2. Wait a few minutes until all items show tick marks, indicating completion, before moving forward.



Step 7: MySqL Installation

1. The downloaded components will be installed.
2. Click "Execute" to start the installation process. MySQL will be installed on your Windows system.
3. Then click Next to proceed.

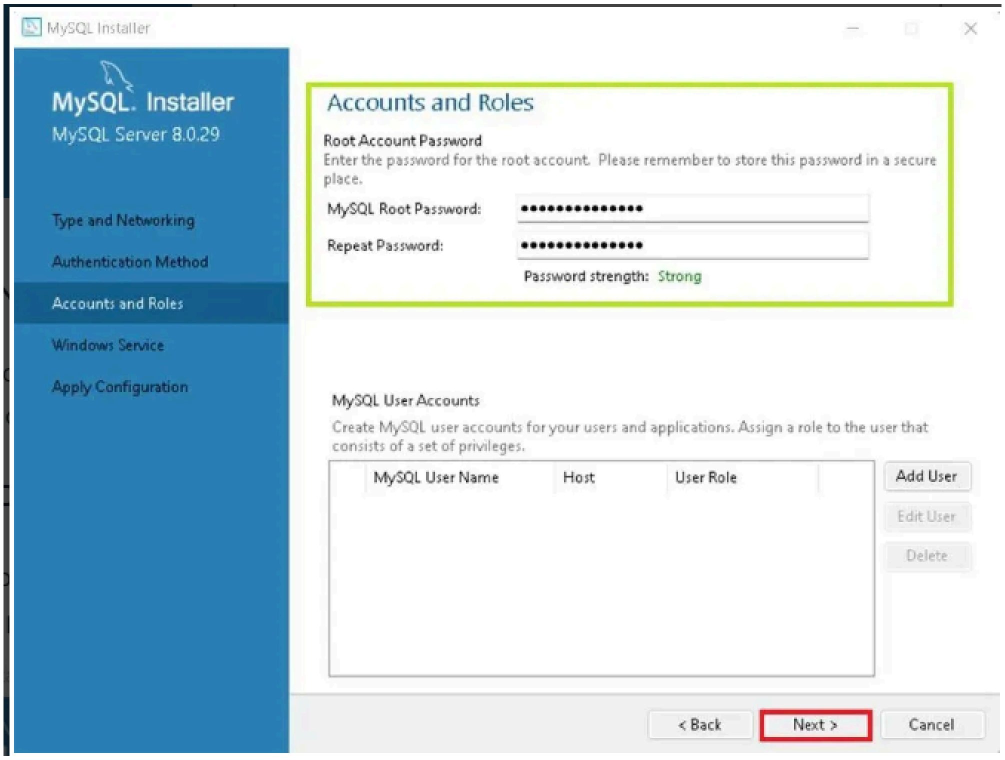


Step 8: Navigate to Few Configuration Pages

1. Proceed to "Product Configuration" > "Type and Networking" > "Authentication Method" Pages by clicking the "Next" button.

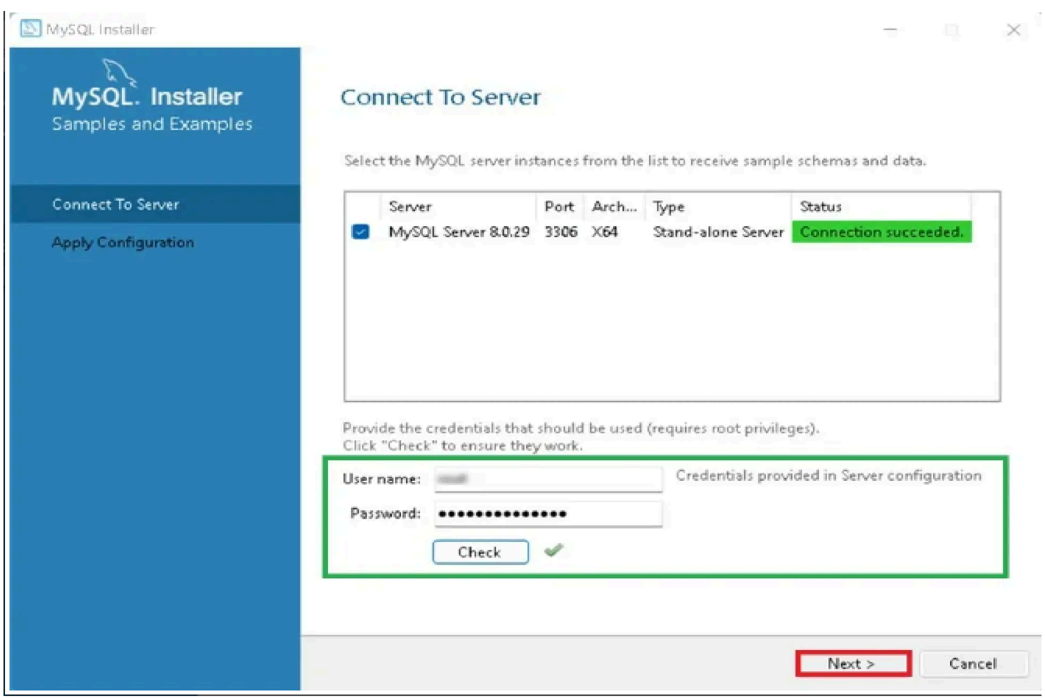
Step 9: Create MySQL Accounts

1. Create a password for the MySQL root user. Ensure it's strong and memorable.
2. Click "Next" to proceed.



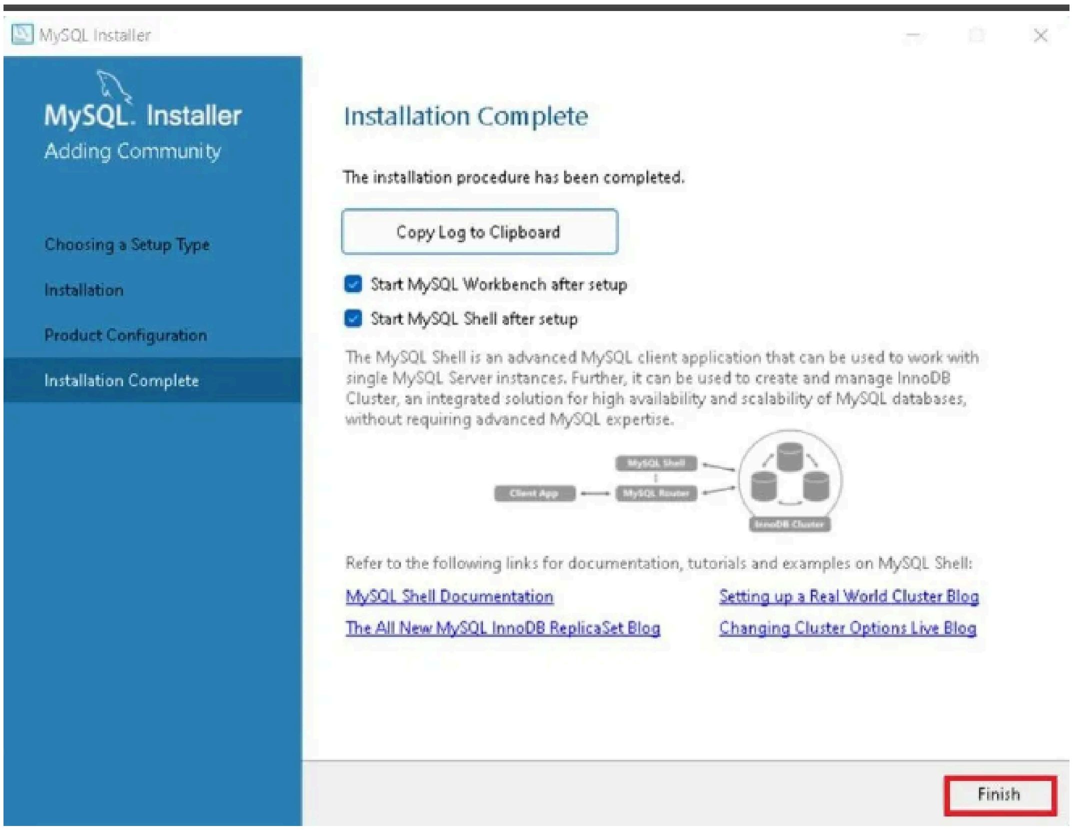
Step 10: Connect To Server

1. Enter the root password, click Check.
2. If it says "Connection succeed," you have successfully connected to the server.



Step 11: Complete Installation

1. Once the installation is complete, click "Finish."
2. MySQL is now installed on your Windows system.



Step 12: Verify Installation

1. To ensure a successful installation of MySQL, open the MySQL Command Line Client or MySQL Workbench, both available in your Start Menu.
2. Log in using the root user credentials you set during installation.